

Task 3: Login System with Validation

Aim: To design and develop a login system that validates user inputs using JavaScript, verifies user credentials from a database, and displays appropriate error messages dynamically.

Algorithm:

1. Start the program
2. Display the login form
3. Enter username and password
4. Validate input using JavaScript (check empty fields)
5. Send data to server (PHP)
6. Connect to database
7. Check credentials using SQL query
8. If valid → show login success
9. Else → show error message
10. Stop the program

Program:

Login.html:

```
<!DOCTYPE html>

<html>

<head>

  <title>Login</title>

  <style>

    body { font-family: Arial; background: #f2f2f2; }

    .box {

      width: 300px;

      margin: 100px auto;

      background: white;

      padding: 20px;

    }

    input {
```

```
    width: 100%;
    padding: 10px;
    margin: 8px 0;
}

button {
    width: 100%;
    padding: 10px;
    background: green;
    color: white;
    border: none;
}

.error { color: red; }
</style>

<script>
    function validateForm() {
        let u = document.forms["f"]["username"].value;
        let p = document.forms["f"]["password"].value;
        if (u == "" || p == "") {
            document.getElementById("error").innerHTML = "All fields required!";
            return false;
        }
        if (p.length < 4) {
            document.getElementById("error").innerHTML = "Password too short!";
            return false;
        }
        return true;
    }
</script>
</head>
```

```

<body>
<div class="box">
    <h2>Login</h2>
    <form name="f" action="login.php" method="POST" onsubmit="return validateForm()">
        <input type="text" name="username" placeholder="Username">
        <input type="password" name="password" placeholder="Password">
        <div id="error" class="error"></div>
        <button type="submit">Login</button>
    </form>
</div>
</body>
</html>

```

Login.php:

```

<?php
include 'db.php';

$username = $_POST['username'];
$password = $_POST['password'];

$sql = "SELECT * FROM users WHERE username='$username' AND
password='$password'";

$result = $conn->query($sql);

if ($result->num_rows > 0) {
    header("Location: home.php");
} else {
    echo "<h3 style='color:red;'>Invalid Username or Password</h3>";
    echo "<a href='login.html'>Try Again</a>";
}

$conn->close();
?>

```

Home.php:

```
<!DOCTYPE html>

<html>

<head>

    <title>Home</title>

</head>

<body>

<h2>Welcome! Login Successful  </h2>

<a href="login.html">Logout</a>

</body>

</html>
```

Database:

```
CREATE DATABASE student_db;

USE student_db;

CREATE TABLE users (

    id INT AUTO_INCREMENT PRIMARY KEY,

    username VARCHAR(50),

    password VARCHAR(50)

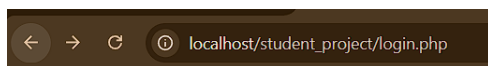
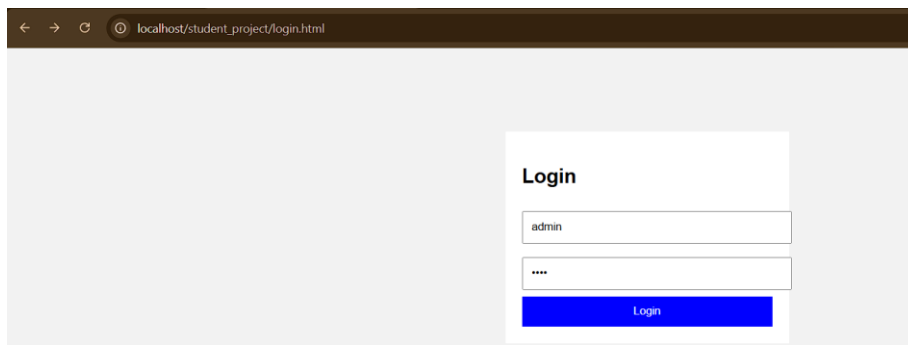
);

INSERT INTO users (username, password) VALUES

('admin', '1234'),

('user1', 'pass123');
```

Output:



Result: Thus, the login page and the connection to the dashboard has been successfully executed and output generated.